

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Agenda 13/30

Session 5. Introduction to contiguously sequential data structures: C-Style arrays, C++ vectors, C++ standard arrays:

Part 2

- More on vectors
- Vector manipulation: writing simple programs
- C-Style arrays vs. C++ vectors
- Introduction to C++ arrays: The array containers
- References and further readings
- Q & A

120 min (incl. Q & A)



Vector_{cont.} and algorithms

vector:

- push_back()
- size()
- subscription operator: []
- empty()
- begin()
- end()

public interface

algorithms:

- sort()
- count()
- find()

...

generic algorithms



- Read some numbers as daily temperatures and compute average and median of them.

```
#include <iostream>
#include <vector>
#include <algorithm>

using std::cin;      using std::cout; using std::endl;
using std::vector; using std::sort;

int main()
{
    vector<double> temps; // temperatures in Fahrenheit, e.g. 64.6
    // to be continued on next page
```

Simple computations, simple programs

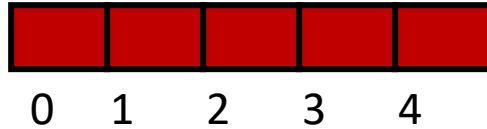
```
// continued from last page
double temp;
while (cin >> temp)
    temps.push_back(temp); // put into vector
double sum = 0;
for (int i = 0; i < temps.size(); ++i)
    sum += temps[i];
cout << "Average temperature: " << sum / temps.size() << endl;
sort(temps.begin(), temps.end()); // sort "from the beginning to the end"

cout<< "Median temperature: " << temps[temps.size() / 2] << endl;
return 0;
}
```

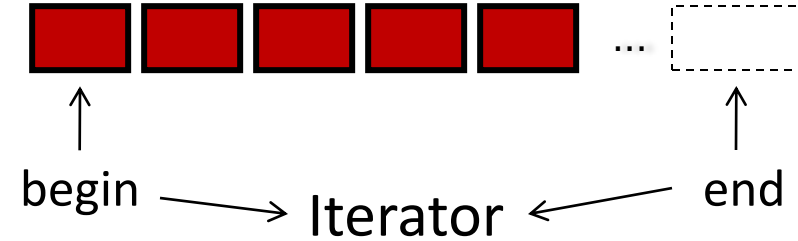
Program

Array vs. vector

C/C++ array



C++ vector



- An array doesn't know its own size

But

- The size of array should be known at compile-time

But

- There is no grow or shrink for arrays.

But

- The name of an array converts to a pointer to its first element.

But

- A vector knows its own size: `size()`

- The size of vector can change at run-time

- Vector can grow or shrink dynamically.

- The name of vector doesn't decay to the first element of container.

Array vs. vector 2.

- Array uses stack or automatic storage
- Array uses pointer arithmetic for traversal.

But

- Vector uses heap or dynamic storage

But

- Vector uses random-access iterator for traversal.

Array container

C-Style Array

Pointer arithmetic	Compile-time static size
Size unaware	automatic storage
Implicit decay name to pointer	No dynamic resize

Array container

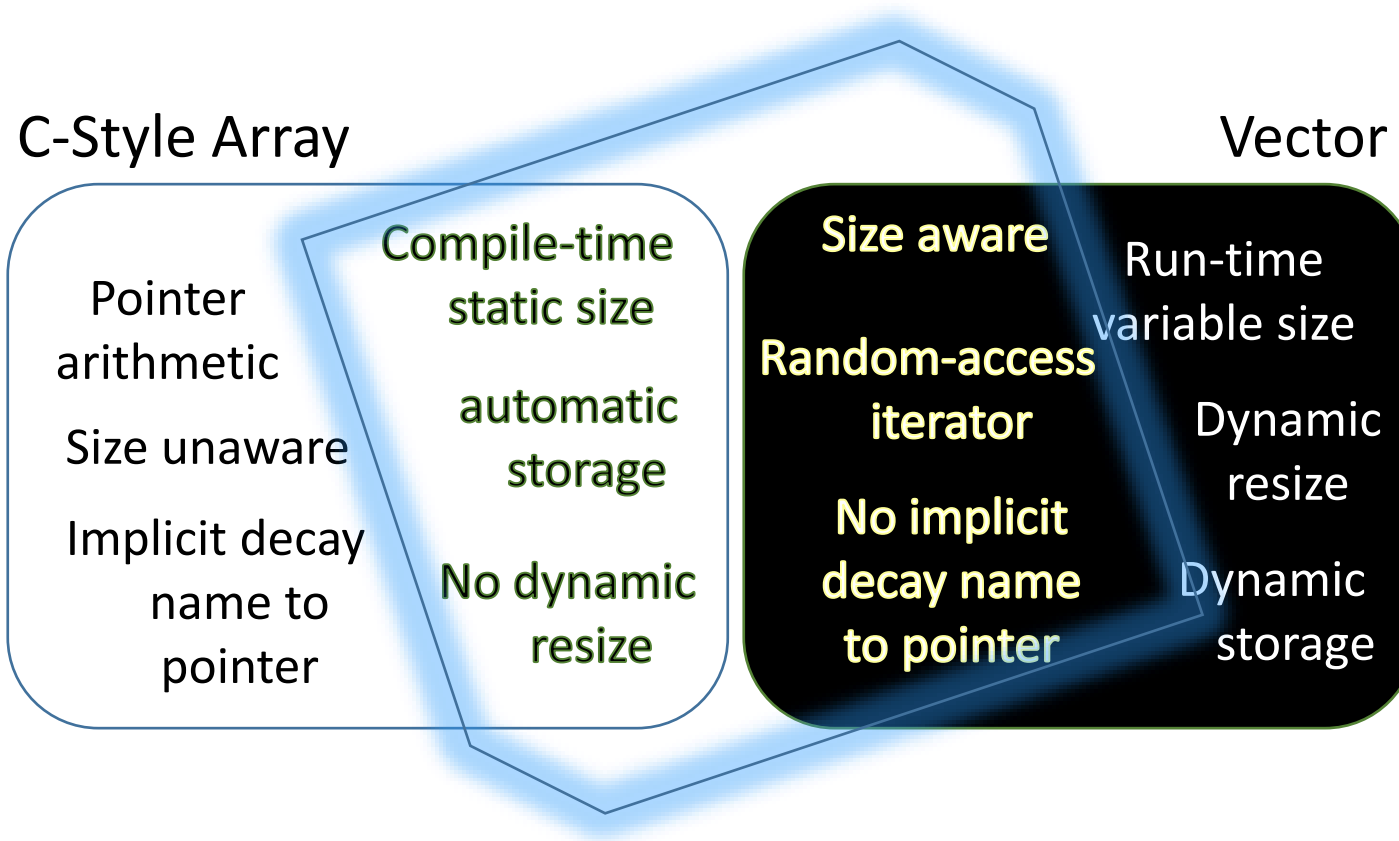
C-Style Array

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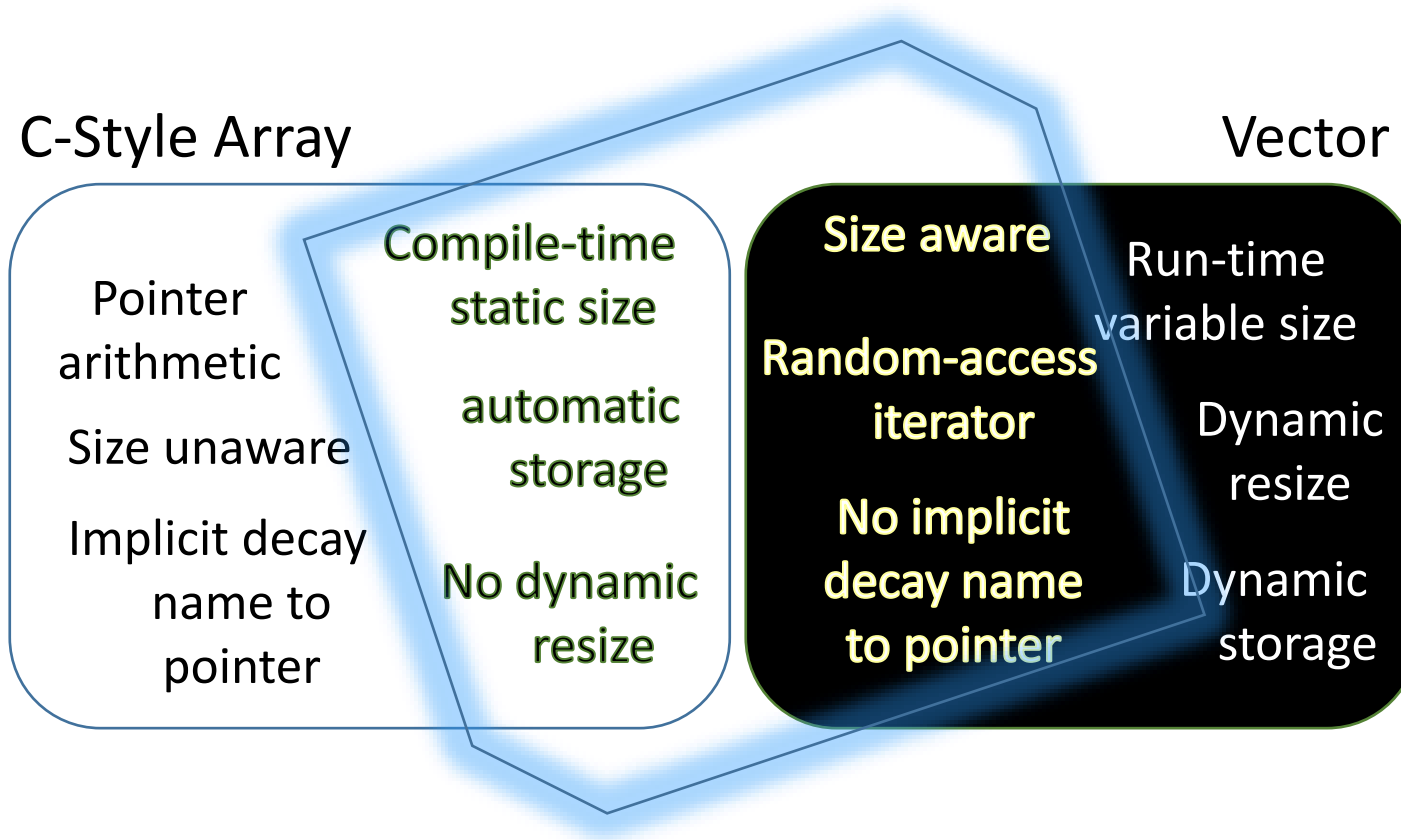
Vector

Size aware	Run-time variable size
Random-access iterator	Dynamic resize
No implicit decay name to pointer	Dynamic storage

Array container



Array container



C++11 standard Array container

C-style array of chars vs. standard library string : comparison

- char string is defined in core language, but string is defined in standard library.
- The size of char string can be determined using null character/strlen() function, but the size of string can be determined using size() member function.
- char string is not generic, but standard string are generic.
- In array of char, we may need to pointer arithmetic but in string we don't.
- In array of char, we may need to memory management but in string we don't.



Use string rather than zero-terminated arrays of char.

R references and further readings

[Str00] Bjarne Stroustrup. The C++ Programming Language (Special Edition). Addison Wesley. Reading Mass. USA. February 2000.

[Str08] Bjarne Stroustrup. Programming: Principles and Practice in C++, 1st edition, Addison Wesley, December 2008.

[Str22] Bjarne Stroustrup. A Tour of C++, 3rd edition, Addison Wesley, September 2022.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

